

2520030453

K.Siri Chandana

Section 6

1.

```
siri_chandana@LAPTOP-MMV  ×  +  ▾
GNU nano 8.7.1 processconcteps1.c
#include <stdio.h>
#include <unistd.h>

int main()
{
    printf("My Process ID (PID) = %d\n", getpid());
    return 0;
}
```

```
siri_chandana@LAPTOP-MMV  ×  +  ▾
siri_chandana@LAPTOP-MMVUDEP:~$ ls
Certifications  Practical  input.txt  practical2.c  processconcepts3.c  processconcepts6.c  processid.c  week1-3.c  week1-6.c  week1-9.c
Ex1             ex.c      output.txt  practical3.c  processconcepts4.c  processconcteps1.c  shell        week1-4.c  week1-7.c  week1.c
OS1.txt.save    filecopy  practical1.c practical4.c  processconcepts5.c  processid          week1-2.c    week1-5.c  week1-8.c
siri_chandana@LAPTOP-MMVUDEP:~$ nano processconcteps1.c
siri_chandana@LAPTOP-MMVUDEP:~$ gcc processconcteps1.c -o shell
siri_chandana@LAPTOP-MMVUDEP:~$ ./shell
My Process ID (PID) = 795
siri_chandana@LAPTOP-MMVUDEP:~$ |
```

2.

```
siri_chandana@LAPTOP-MMV  ×  +  ▾
GNU nano 8.7.1 processid.c
#include <stdio.h>
#include <unistd.h>
int main() {
    pid_t pid=fork();
    if(pid==0)
    {
        printf("Child Process\n");
        printf("PID = %d\n" , getpid());
    }
    else
    {
        printf("Parent Process\n");
        printf("PID = %d\n",getpid());
    }
}
```

```
siri_chandana@LAPTOP-MMV x + v
siri_chandana@LAPTOP-MMVUDEP:~$ ls
Certifications Practical input.txt practical2.c processconcepts3.c processconcepts6.c processid.c week1-3.c week1-6.c week1-9.c
Ex1 ex.c output.txt practical3.c processconcepts4.c processconcepts1.c shell week1-4.c week1-7.c week1.c
OS1.txt.save filecopy practical1.c practical4.c processconcepts5.c processid week1-2.c week1-5.c week1-8.c
siri_chandana@LAPTOP-MMVUDEP:~$ nano processconcepts1.c
siri_chandana@LAPTOP-MMVUDEP:~$ gcc processconcepts1.c -o shell
siri_chandana@LAPTOP-MMVUDEP:~$ ./shell
My Process ID (PID) = 795
siri_chandana@LAPTOP-MMVUDEP:~$ nano processid.c
siri_chandana@LAPTOP-MMVUDEP:~$ gcc processid.c -o shell
siri_chandana@LAPTOP-MMVUDEP:~$ ./shell
Parent Process
PID = 835
Child Process
PID = 836
siri_chandana@LAPTOP-MMVUDEP:~$ |
```

3.

```
siri_chandana@LAPTOP-MMV x + v
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
int main() {
pid_t pid=fork();
if(pid==0)
{
printf("Child Process Executing...\n");
sleep(3);
printf("Child Finished\n");
}
else
{
wait(NULL);
printf("Parent Process Continues\n");
}
return 0;
}
```

```
siri_chandana@LAPTOP-MMV x + v
siri_chandana@LAPTOP-MMVUDEP:~$ ls
Certifications Practical input.txt practical2.c processconcepts3.c processconcepts6.c processid.c week1-3.c week1-6.c week1-9.c
Ex1 ex.c output.txt practical3.c processconcepts4.c processconcepts1.c shell week1-4.c week1-7.c week1.c
OS1.txt.save filecopy practical1.c practical4.c processconcepts5.c processid week1-2.c week1-5.c week1-8.c
siri_chandana@LAPTOP-MMVUDEP:~$ nano processconcepts1.c
siri_chandana@LAPTOP-MMVUDEP:~$ gcc processconcepts1.c -o shell
siri_chandana@LAPTOP-MMVUDEP:~$ ./shell
My Process ID (PID) = 795
siri_chandana@LAPTOP-MMVUDEP:~$ nano processid.c
siri_chandana@LAPTOP-MMVUDEP:~$ gcc processid.c -o shell
siri_chandana@LAPTOP-MMVUDEP:~$ ./shell
Parent Process
PID = 835
Child Process
PID = 836
siri_chandana@LAPTOP-MMVUDEP:~$ nano processconcepts3.c
siri_chandana@LAPTOP-MMVUDEP:~$ gcc processconcepts3.c -o shell
siri_chandana@LAPTOP-MMVUDEP:~$ ./shell
Child Process Executing...
Child Finished
Parent Process Continues
siri_chandana@LAPTOP-MMVUDEP:~$ |
```

4.

```
siri_chandana@LAPTOP-MMV X + v
GNU nano 8.7.1 processconcepts4.c
#include <stdio.h>
#include <unistd.h>

int main()
{
    for(int i = 1; i <= 3; i++)
    {
        if(fork() == 0)
        {
            printf("Child %d PID = %d\n", i, getpid());
            return 0;
        }
    }
    return 0;
}
```

```
siri_chandana@LAPTOP-MMV X + v
siri_chandana@LAPTOP-MMVUEP:~$ ls
Certifications Practical input.txt practical2.c processconcepts3.c processconcepts6.c processid.c week1-3.c week1-6.c week1-9.c
Ex1 ex.c output.txt practical3.c processconcepts4.c processconcepts1.c shell week1-4.c week1-7.c week1.c
OS1.txt.save filecopy practical1.c practical4.c processconcepts5.c processid week1-2.c week1-5.c week1-8.c
siri_chandana@LAPTOP-MMVUEP:~$ nano processconcepts1.c
siri_chandana@LAPTOP-MMVUEP:~$ gcc processconcepts1.c -o shell
siri_chandana@LAPTOP-MMVUEP:~$ ./shell
My Process ID (PID) = 795
siri_chandana@LAPTOP-MMVUEP:~$ nano processid.c
siri_chandana@LAPTOP-MMVUEP:~$ gcc processid.c -o shell
siri_chandana@LAPTOP-MMVUEP:~$ ./shell
Parent Process
PID = 835
Child Process
PID = 836
siri_chandana@LAPTOP-MMVUEP:~$ nano processconcepts3.c
siri_chandana@LAPTOP-MMVUEP:~$ gcc processconcepts3.c -o shell
siri_chandana@LAPTOP-MMVUEP:~$ ./shell
Child Process Executing...
Child Finished
Parent Process Continues
siri_chandana@LAPTOP-MMVUEP:~$ nano processconcepts4.c
siri_chandana@LAPTOP-MMVUEP:~$ gcc processconcepts4.c -o shell
siri_chandana@LAPTOP-MMVUEP:~$ ./shell
Child 1 PID = 920
Child 2 PID = 921
Child 3 PID = 922
siri_chandana@LAPTOP-MMVUEP:~$ |
```

5.

```
siri_chandana@LAPTOP-MMV X + v
GNU nano 8.7.1 processconcepts5.c
#include <stdio.h>
#include <unistd.h>
int main()
{
    printf("Before exec()\n");
    execlp("ls", "ls", "-l", NULL);
    printf("This will not execute.\n");
    return 0;
}
```

```
siri_chandana@LAPTOP-MMV  x  +  v
siri_chandana@LAPTOP-MMVUDEP:~$ nano processconcepts4.c
siri_chandana@LAPTOP-MMVUDEP:~$ gcc processconcepts4.c -o shell
siri_chandana@LAPTOP-MMVUDEP:~$ ./shell
Child 1 PID = 920
Child 2 PID = 921
Child 3 PID = 922
siri_chandana@LAPTOP-MMVUDEP:~$ nano processconcepts5.c
siri_chandana@LAPTOP-MMVUDEP:~$ gcc processconcepts5.c -o shell
siri_chandana@LAPTOP-MMVUDEP:~$ ./shell
Before exec()
total 152
drwxr-xr-x 2 siri_chandana siri_chandana 4096 Aug  4 06:37 Certifications
drwxr-xr-x 2 siri_chandana siri_chandana 4096 Jul 28 03:59 Ex1
-rw-r--r-- 1 siri_chandana siri_chandana  5 Jul 31 09:12 OS1.txt.save
drwxr-xr-x 3 siri_chandana siri_chandana 4096 Aug  4 04:04 Practical
-rw-r--r-- 1 siri_chandana siri_chandana 123 Aug  4 03:44 ex.c
-rwxr-xr-x 1 siri_chandana siri_chandana 16176 Aug  7 09:39 filecopy
-rw-r--r-- 1 siri_chandana siri_chandana  79 Aug  7 09:37 input.txt
-rw-r--r-- 1 siri_chandana siri_chandana  79 Aug  7 09:39 output.txt
-rw-r--r-- 1 siri_chandana siri_chandana  562 Aug  7 09:31 practical1.c
-rw-r--r-- 1 siri_chandana siri_chandana  534 Aug  7 09:35 practical2.c
-rw-r--r-- 1 siri_chandana siri_chandana  584 Aug 15 12:02 practical3.c
-rw-r--r-- 1 siri_chandana siri_chandana  407 Aug 15 12:04 practical4.c
-rw-r--r-- 1 siri_chandana siri_chandana  256 Aug 15 12:13 processconcepts3.c
-rw-r--r-- 1 siri_chandana siri_chandana  240 Aug 15 12:14 processconcepts4.c
-rw-r--r-- 1 siri_chandana siri_chandana  156 Aug 15 12:15 processconcepts5.c
-rw-r--r-- 1 siri_chandana siri_chandana  179 Aug 14 10:17 processconcepts6.c
-rw-r--r-- 1 siri_chandana siri_chandana  121 Aug 15 12:10 processconcepts1.c
-rwxr-xr-x 1 siri_chandana siri_chandana 16080 Aug 14 09:49 processid
-rw-r--r-- 1 siri_chandana siri_chandana  215 Aug 15 12:11 processid.c
-rwxr-xr-x 1 siri_chandana siri_chandana 16008 Aug 15 12:15 shell
-rw-r--r-- 1 siri_chandana siri_chandana  158 Jul 31 09:56 week1-2.c
-rw-r--r-- 1 siri_chandana siri_chandana  110 Jul 31 10:01 week1-3.c
-rw-r--r-- 1 siri_chandana siri_chandana  104 Jul 31 10:03 week1-4.c
-rw-r--r-- 1 siri_chandana siri_chandana  230 Jul 31 10:04 week1-5.c
-rw-r--r-- 1 siri_chandana siri_chandana  287 Jul 31 10:05 week1-6.c
-rw-r--r-- 1 siri_chandana siri_chandana  274 Jul 31 10:06 week1-7.c
-rw-r--r-- 1 siri_chandana siri_chandana  159 Jul 31 10:07 week1-8.c
-rw-r--r-- 1 siri_chandana siri_chandana  264 Jul 31 10:08 week1-9.c
-rw-r--r-- 1 siri_chandana siri_chandana   78 Jul 31 09:15 week1.c
siri_chandana@LAPTOP-MMVUDEP:~$ |
```

6.

```
siri_chandana@LAPTOP-MMV  x  +  v
GNU nano 8.7.1 processconcepts6.c
#include <stdio.h>
#include <unistd.h>

int main()
{
    printf("Before exec()\n");

    execlp("ls", "ls", "-l", NULL);

    printf("This will not execute.\n");

    return 0;
}
```

```

-rw-r--r-- 1 siri_chandana siri_chandana 78 Jul 31 09:15 week1.c
siri_chandana@LAPTOP-MMVMUDEP:~$ nano processconcepts6.c
siri_chandana@LAPTOP-MMVMUDEP:~$ gcc processconcepts6.c -o shell
siri_chandana@LAPTOP-MMVMUDEP:~$ ./shell
Before exec()
total 152
drwxr-xr-x 2 siri_chandana siri_chandana 4096 Aug 4 06:37 Certifications
drwxr-xr-x 2 siri_chandana siri_chandana 4096 Jul 28 03:59 Ex1
-rw----- 1 siri_chandana siri_chandana 5 Jul 31 09:12 OS1.txt.save
drwxr-xr-x 3 siri_chandana siri_chandana 4096 Aug 4 04:04 Practical
-rw-r--r-- 1 siri_chandana siri_chandana 123 Aug 4 03:44 ex.c
-rwxr-xr-x 1 siri_chandana siri_chandana 16176 Aug 7 09:39 filecopy
-rw-r--r-- 1 siri_chandana siri_chandana 79 Aug 7 09:37 input.txt
-rw-r--r-- 1 siri_chandana siri_chandana 79 Aug 7 09:39 output.txt
-rw-r--r-- 1 siri_chandana siri_chandana 562 Aug 7 09:31 practical1.c
-rw-r--r-- 1 siri_chandana siri_chandana 534 Aug 7 09:35 practical2.c
-rw-r--r-- 1 siri_chandana siri_chandana 584 Aug 15 12:02 practical3.c
-rw-r--r-- 1 siri_chandana siri_chandana 407 Aug 15 12:04 practical4.c
-rw-r--r-- 1 siri_chandana siri_chandana 256 Aug 15 12:13 processconcepts3.c
-rw-r--r-- 1 siri_chandana siri_chandana 240 Aug 15 12:14 processconcepts4.c
-rw-r--r-- 1 siri_chandana siri_chandana 156 Aug 15 12:15 processconcepts5.c
-rw-r--r-- 1 siri_chandana siri_chandana 179 Aug 15 12:16 processconcepts6.c
-rw-r--r-- 1 siri_chandana siri_chandana 121 Aug 15 12:10 processconcepts1.c
-rwxr-xr-x 1 siri_chandana siri_chandana 16080 Aug 14 09:49 processid
-rw-r--r-- 1 siri_chandana siri_chandana 215 Aug 15 12:11 processid.c
-rwxr-xr-x 1 siri_chandana siri_chandana 16008 Aug 15 12:16 shell
-rw-r--r-- 1 siri_chandana siri_chandana 158 Jul 31 09:56 week1-2.c
-rw-r--r-- 1 siri_chandana siri_chandana 110 Jul 31 10:01 week1-3.c
-rw-r--r-- 1 siri_chandana siri_chandana 104 Jul 31 10:03 week1-4.c
-rw-r--r-- 1 siri_chandana siri_chandana 230 Jul 31 10:04 week1-5.c
-rw-r--r-- 1 siri_chandana siri_chandana 287 Jul 31 10:05 week1-6.c
-rw-r--r-- 1 siri_chandana siri_chandana 274 Jul 31 10:06 week1-7.c
-rw-r--r-- 1 siri_chandana siri_chandana 159 Jul 31 10:07 week1-8.c
-rw-r--r-- 1 siri_chandana siri_chandana 264 Jul 31 10:08 week1-9.c
-rw-r--r-- 1 siri_chandana siri_chandana 78 Jul 31 09:15 week1.c
siri_chandana@LAPTOP-MMVMUDEP:~$ |

```